CHAPTER 14

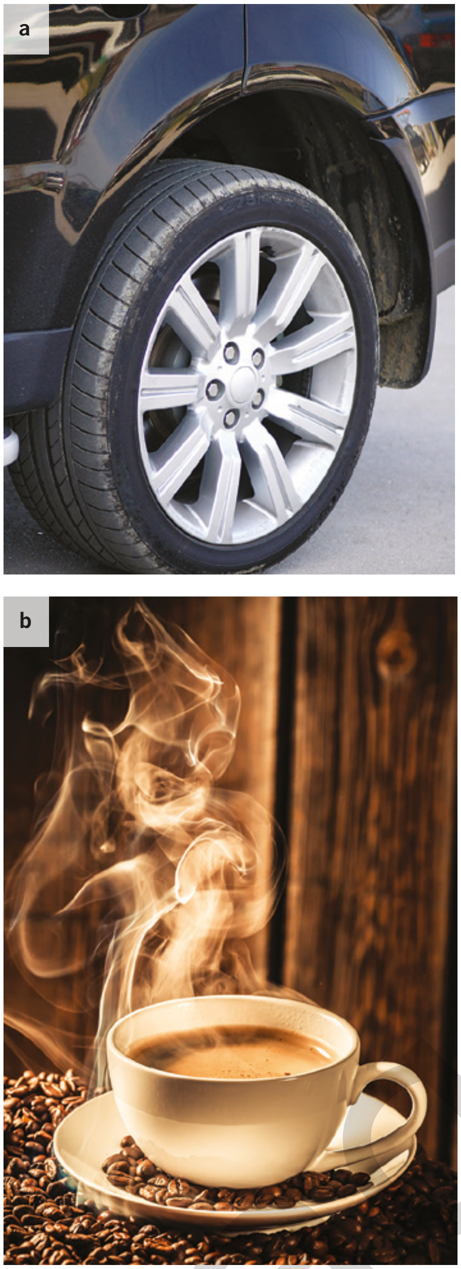
Gases

At the end of this chapter, you will be able to describe the properties of gases and relate these to the kinetic molecular theory. You will understand how the pressure and volume of a gas sample are measured and explain the relationship between volume, pressure and temperature. You will learn how the volume of a gas can be calculated from its amount, measured in moles.

Science understanding

- the behaviour of an ideal gas, including the qualitative relationships between pressure, temperature and volume, can be explained using the Kinetic Theory
- the mole concept can be used to calculate the mass of substances and volume of gases (at standard temperature and pressure) involved in a chemical reaction

14.1 Introducing gases



There are countless everyday examples you observe of the behaviour of gases (Figure 14.1.1). Such examples tell us a great deal about the physical properties of gases—those properties that can be observed and measured without changing the nature of the gas itself.

In this section, you will learn about the properties and behaviour of gases.

PROPERTIES OF GASES

Table 14.1.1 summarises some of the properties of gases and compares them with the properties of solids and liquids. These observations can be used to develop a particle model of gas behaviour.

TABLE 14.1.1 Some properties of the three states of matter

	Gases	Liquids	Solids
Density	low density	high density	high density
Volume and shape	fill the space available, because particles move independently of one another	fixed volume, adopt shape of container, because particles are affected by attractive forces	fixed volume and shape, because particles are affected by attractive forces
Compressibility	compress easily	almost incompressible	almost incompressible
Ability to mix	gases rapidly mix together	liquids slowly mix together unless stirred	solids do not mix unless finely divided

The low density of gases, relative to that of liquids and solids, suggests that the particles in a gas are much more widely spaced. The mass of any gas in a given **volume** will be less than the mass of a liquid or solid in the same volume. This is consistent with the observation that gases are easily compressed—a property fundamental to scuba diving, as seen in Figure 14.1.2. The air that the scuba diver breathes must be compressed to fit into a scuba tank carried on the diver's back.

FIGURE 14.1.1 (a) Air is used to inflate vehicle tyres. Air is a mixture of gases and is easily compressed. When the car goes over a bump in the road, the air compresses slightly and absorbs the impact of the bump. (b) The gases that cause the smell of a freshly brewed cup of coffee rapidly fill an entire room. Gases mix readily and, unlike solids and liquids, occupy all available space.

FIGURE 14.1.2 The underwater activity of a scuba diver relies on the physical properties of gases, including their ability to be compressed.

Gases spread to fill the space available, as shown in Figure 14.1.3. This suggests that the particles of a gas move independently of each other.

The wide spacing and independent movement of particles explains why different gases mix rapidly.

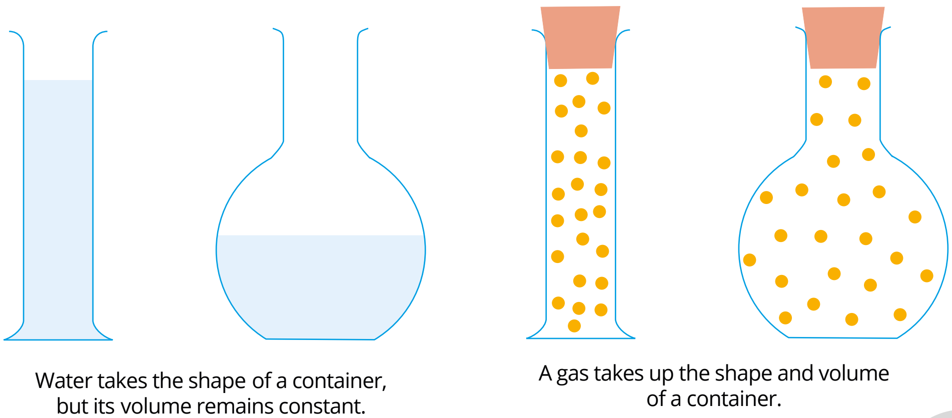


FIGURE 14.1.3 Both liquids and gases take the shape of the container they are in. However, a liquid has a fixed volume, whereas the volume of a gas expands to fill all available space in a container.

KINETIC MOLECULAR THEORY

Scientists have developed a model to explain gas behaviour based on the behaviour of the particles of a gas. This model is known as the **kinetic molecular theory** of gases. The theory is based on the following assumptions:

- Gases are composed of small particles, either atoms or molecules.
- The total volume occupied by the particles in a gas is negligible compared to the average distance between them. Consequently, most of the volume occupied by a gas is empty space.
- Gas particles move rapidly in random, straight-line motion.
- Particles collide with each other and with the walls of the container.
- The forces between particles are extremely weak.
- **Kinetic energy**, the energy of motion, can be transferred from one particle to another, but the total kinetic energy remains constant. Therefore, collisions between gas particles are **elastic collisions**—kinetic energy is conserved.
- The average kinetic energy of the particles increases as the temperature of the gas increases.

Ideal gas behaviour

A gas that conforms to the criteria of the kinetic molecular theory of gases is described as behaving as an **ideal gas**. In particular, in an ideal gas the total volume occupied by the particles in a gas is negligible compared to the total volume of the gas and collisions between gas particles are elastic. Ideal gas behaviour allows us to predict the behaviour of gases in terms of how they respond to changes in pressure, volume or temperature.

Most gases that you encounter, such as oxygen, hydrogen and carbon dioxide, behave as ideal gases, especially at high temperatures and low pressures, when the space between the atoms or molecules of gas is maximised. As gases cool down, or are placed under higher pressures, the particles come closer together and the increased interactions between the particles cause the gases to show non-ideal behaviour. Likewise, gases that contain molecules with higher relative molecular masses will not behave ideally due to the larger size of the particles in the gas.

CHEMFILE

Airbags in cars

Front and side airbags are now standard equipment in new cars. Airbags contain a mixture of crystalline solids—sodium azide (NaN_3), potassium nitrate (KNO_3) and silica (SiO_2)—stored in a canister. When there is an accident, a sensor ‘ignites’ the sodium azide, initiating the first step in this reaction. Sodium metal and hot nitrogen gas are products of this energy-releasing reaction. The nitrogen gas rapidly expands, which inflates the nylon airbag. In the second part of the reaction, the molten sodium metal immediately reacts with the potassium nitrate, generating more nitrogen gas, as well as sodium oxide and potassium oxide, which are white powdery solids.

FIGURE 14.1.4 An airbag deploys in a test car within 30 milliseconds of impact, with a crash dummy taking the impact.

THE RELATIONSHIP BETWEEN MOLECULAR KINETIC ENERGY AND TEMPERATURE

According to the kinetic molecular theory, the average kinetic energy of gas particles in a sample is directly proportional to the temperature of the gas. This means that as the temperature of a gas sample increases, the average kinetic energy of the particles in the gas also increases. However, it is important to understand that at any given temperature:

- the average kinetic energy of a gas does not depend on the chemical identity of the gas. For example, molecules of hydrogen gas and molecules of oxygen gas will have the same average kinetic energy at the same temperature.
- not all particles within a sample will have the same kinetic energy. Some particles will have a lower kinetic energy and some will have a higher kinetic energy. The temperature reflects the average kinetic energy of all particles in the sample.

The distribution of the kinetic energies of particles in a gas at a given temperature is illustrated by the graph in Figure 14.1.5.

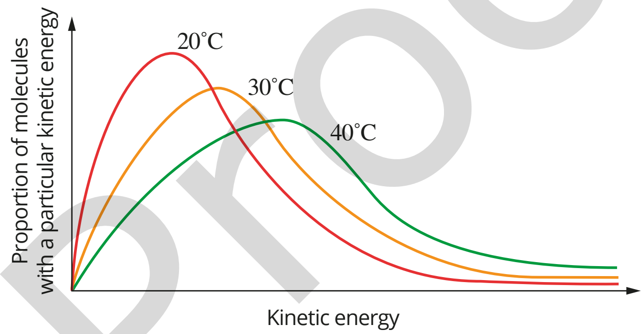


FIGURE 14.1.5 The distribution of molecular kinetic energies for three different temperatures

This graph is called a **Maxwell–Boltzmann distribution graph** and is also known as a **kinetic energy distribution diagram**. The total area under each curve represents the total number of molecules in a given sample. The areas under the three curves are equal, because the same number of molecules is being compared at three different temperatures.

Notice that, at 20°C, a greater proportion of molecules have a lower kinetic energy. The reverse is true at 40°C. At this higher temperature, a greater proportion of molecules have a higher kinetic energy.

From the figure above, you can see that:

- the proportion of molecules with high kinetic energy increases with temperature
- the average kinetic energy of a sample increases with temperature
- only a small proportion of molecules have very low or very high kinetic energy at any temperature.

The average kinetic energy of particles in gases is related to their average speed. Recall that different gases at the same temperature have the same average kinetic energy and that kinetic energy (KE) is given by the formula KE = $\frac{1}{2}mv^2$, where m = mass and v = velocity. Therefore, at a given temperature, particles with a higher mass on average travel at a lower velocity than particles with a lower mass.

Diffusion

Diffusion is the process whereby gases in a mixture spread out to uniformly fill the total volume available (Figure 14.1.6). The rate at which gases diffuse depends on the temperature of the gases, their molar masses, and the average velocity of their particles.

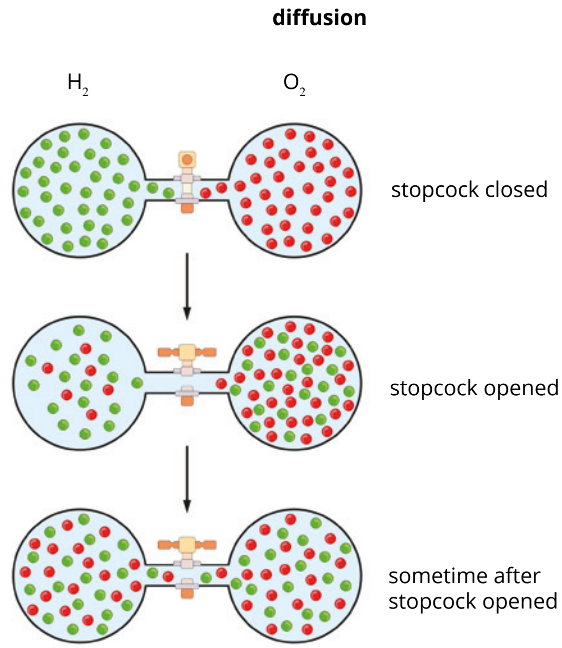


FIGURE 14.1.6 The diffusion of H_2 and O_2 particles in gaseous state when allowed to uniformly spread throughout a container. The gaseous particles will keep moving until a homogeneous mixture is achieved. Note that the hydrogen gas diffuses more quickly into the oxygen than the oxygen gas diffuses into the hydrogen.

You learnt earlier that as temperature increases in a specific sample of gas, the average kinetic energy of the particles also increases. In this case, the mass of the particles is held constant. Therefore, the velocity and speed of the particles increase with temperature.

Conversely, if the temperature of a sample of gases is held constant, the average kinetic energy of particles is constant. In this case, because $KE = \frac{1}{2}mv^2$, the lighter the particles of gas, the greater their velocity. At a given temperature, gases of lower molecular mass move at a greater velocity and, therefore, will diffuse more rapidly than gases of higher molecular mass This is why the hydrogen gas shown in Figure 14.1.6 will diffuse more rapidly than the oxygen gas. Figure 14.1.7 shows an example of diffusion.

FIGURE 14.1.7 White ammonium chloride powder is formed as gases from concentrated ammonia solution and hydrochloric acid solution diffuse from their containers and react. Ammonia and hydrogen chloride are colourless gases.

THE NATURE OF VOLUME AND PRESSURE

When you blow up a balloon, the balloon expands because you put more air into it and the rubber of the balloon stretches. If you then plunge that balloon into liquid nitrogen, as shown in Figure 14.1.8, the balloon shrinks dramatically. The temperature of the gas decreases, causing the volume to decrease.

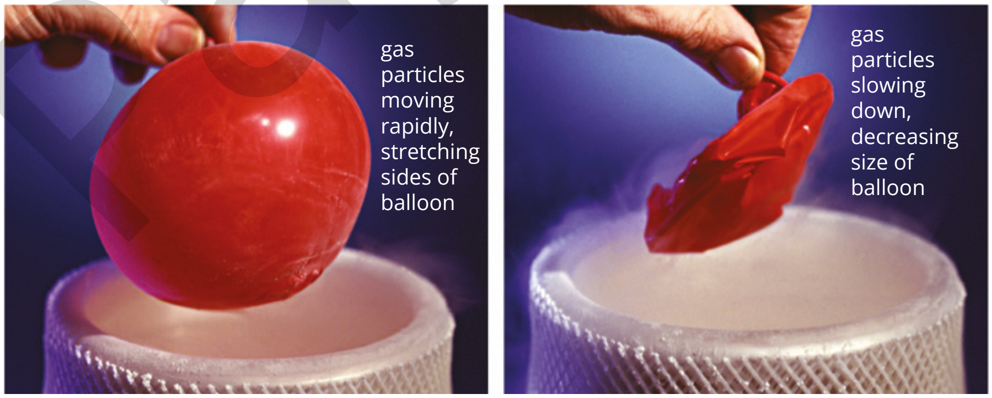


FIGURE 14.1.8 A balloon filled with air at room temperature is dipped into liquid nitrogen at −196°C. The volume of the air inside the balloon decreases dramatically.

FIGURE 14.1.9 Pumping more air into a tyre increases the pressure in the tyre because more particles are being pumped into a nearly fixed volume.

Pumping more air into a tyre, as shown in Figure 14.1.9, increases the pressure inside the tyre because more particles are being forced into approximately the same volume inside the tyre.

Volume

Volume is the quantity that is used to describe the space that a substance occupies. Because a gas occupies the whole container that it is in, the volume of a gas is equal to the volume of its container.

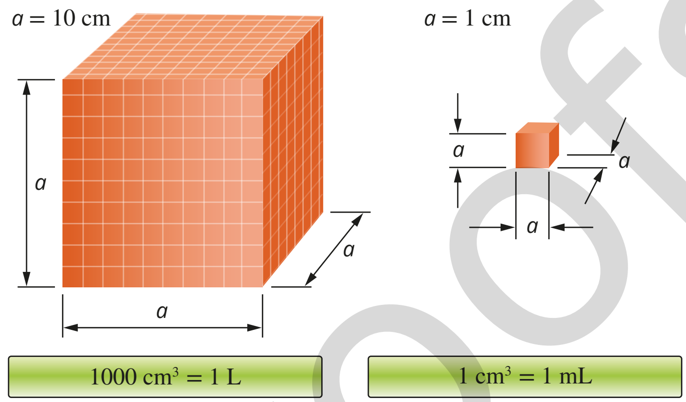


FIGURE 14.1.10 Cubic centimetres, millilitres and litres are the most commonly used units of volume.

There are several different units used for volume, some of which are represented in Figure 14.1.10. The common units are litre (L), millilitre (mL), cubic metre (m^3) and cubic centimetre (cm^3).

- $1\,mL = 1\,cm^3$
- $1\,L = 1 \times 10^3\,mL$ or $1000\,mL$
- $1\,m^3 = 1000\,L = 1 \times 10^6\,mL$

Small volumes of gas are usually measured in millilitres (mL) or litres (L). Very large samples are measured in cubic metres (m^3).

Different units are used to describe volume. The relationship between these units is:

$1000\,mL = 1000\,cm^3 = 1\,L$

$10^6\,mL = 10^6\,cm^3 = 1\,m^3$

Worked example 14.1.1

CONVERTING VOLUME UNITS

A gas has a volume of 255 mL.
What is its volume in:

a cubic centimetres (cm^3)?

b litres (L)?

c cubic metres (m^3)?

Thinking	Working
Recall the conversion factors for each of the units of volume. Apply the correct conversion to each situation.	**a** The units of mL and cm^3 are equivalent. $1\,mL = 1\,cm^3$ $255\,mL = 255\,cm^3$ **b** $1000\,mL = 1\,L$ Divide volume in mL by 1000 to convert to L. $255\,mL = \frac{255}{1000}$ $= 0.255\,L$ **c** $1 \times 10^6\,mL = 1\,m^3$ Divide volume in mL by 1×10^6 to convert to m^3. $255\,mL = \frac{255}{1 \times 10^6}$ $= 2.55 \times 10^{-4}\,m^3$

Worked example: Try yourself 14.1.1

CONVERTING VOLUME UNITS

A gas has a volume of 700 mL.
What is its volume in:

a cubic centimetres (cm^3)?
b litres (L)?
c cubic metres (m^3)?

Pressure

People often talk about exerting pressure on something as if it is some kind of force. In terms of the kinetic theory of gases, **pressure** is defined as the force exerted on a unit area of a surface by the particles of a gas as they collide with the surface.

The more a gas is compressed, the greater the number of collisions the gas particles will have with each other and the walls of their container. These collisions produce a force on the walls of the container, such as the inside of a tyre, which can be measured. The force per unit area is described as pressure.

The pressure of a gas can be measured in different ways. Figure 14.1.11 shows a pressure gauge like those attached to cylinders of compressed air used for scuba diving.

FIGURE 14.1.11 A pressure gauge like this one would be found on a cylinder of compressed air used in scuba diving.

Air is a mixture of gases including nitrogen, oxygen, carbon dioxide and argon. In a gaseous mixture, such as air, the measured air pressure is the sum of pressures of the individual gases in the mixture (see figure below). In air, the nitrogen molecules collide with the walls of a container, exerting a pressure. In a similar way, the oxygen molecules exert a pressure, as do molecules of each gas present in the mixture. In the gaseous mixture of nitrogen and oxygen shown in Figure 14.1.12, the measured pressure is the sum of the **partial pressure** of oxygen and the partial pressure of nitrogen.

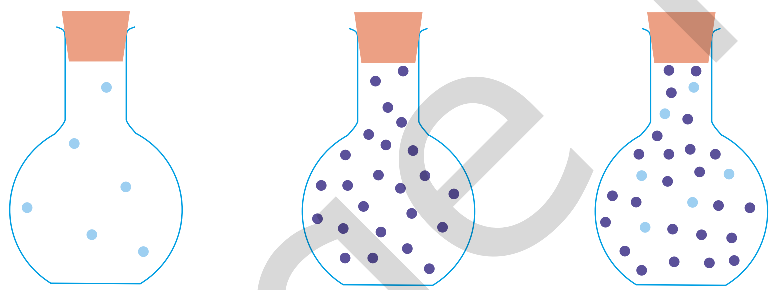


FIGURE 14.1.12 The total pressure of a mixture of gases is the sum of the partial pressures (individual pressures) of each of the gases in the mixture.

Units of pressure

Since pressure is the force exerted on a unit area of a surface, the relationship can be written as:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$P = \frac{F}{A}$$

The units of pressure depend on the units used to measure force and area. Over the years, scientists in different countries have used different units to measure force and area, so there are a number of different units of pressure.

CHEMFILE

Torricelli's barometer

In the 17th century, the Italian physicist Evangelista Torricelli invented the earliest barometer, an instrument used to measure atmospheric pressure. It was a straight glass tube, closed at one end, containing mercury. The tube was inverted so that the open end was below the surface of mercury in a bowl as seen in Figure 14.1.13.

The column of mercury in Torricelli's barometer was supported by the pressure of the gas particles in the atmosphere colliding with the surface of the mercury in the open bowl.

At sea level, the top of the column of mercury was about 760 mm above the surface of the mercury in the bowl. Torricelli found that the height of the mercury column decreased when he took his barometer to higher altitudes in the mountains.

At higher altitudes, there are fewer air particles and therefore less frequent collisions on the surface area of mercury. The reduced pressure supports a shorter column of mercury.

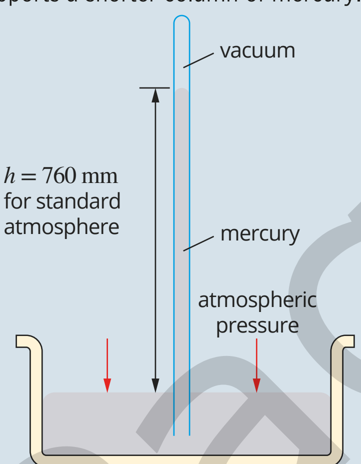


FIGURE 14.1.13 A simple Torricelli barometer

The SI unit for force is the newton (N) and for area is the square metre (m^2). Pressure in SI units is therefore newton per square metre ($N\,m^{-2}$). One newton per square metre is equivalent to a pressure of one **pascal** (Pa).

In 1982, the International Union of Pure and Applied Chemistry (IUPAC), the organisation responsible for naming chemicals and setting standards, adopted a standard for pressure equivalent to 100 000 Pa or 100 kPa. This gave rise to a new pressure unit, the **bar**, where 1 bar equals 100 kPa.

The use of mercury barometers in past years resulted in pressure often being measured in millimetres of mercury, or mmHg.

Another unit for pressure is the standard atmosphere (atm). One standard atmosphere (1 atm) is the pressure required to support 760 millimetres of mercury (760 mmHg) in a mercury barometer at 25°C. This is the average atmospheric pressure at sea level. One atmosphere equals 101.3 kPa.

Summary of pressure units

There are four common units of gas pressure: pascal, bar, millimetres of mercury and atmosphere (Table 14.1.2).

TABLE 14.1.2 Common units of gas pressure

Name of unit	Symbol for unit	Conversion to $N\,m^{-2}$
newton per square metre	$N\,m^{-2}$	
pascal	Pa	$1\,Pa = 1\,N\,m^{-2}$
kilopascal	kPa	$1\,kPa = 1 \times 10^3\,Pa = 1 \times 10^3\,N\,m^{-2}$
atmosphere	atm	$1\,atm = 101.3\,kPa = 1.013 \times 10^5\,N\,m^{-2}$
bar	bar	$1\,bar = 100\,kPa = 1.00 \times 10^5\,N\,m^{-2}$
millimetres of mercury	mmHg	$760\,mmHg = 1\,atm = 1.013 \times 10^5\,N\,m^{-2}$

These relationships can be used to convert pressure from one unit to another.

Different units are used to describe pressure. The relationship between these is:

$1\,bar = 100\,kPa = 1.00 \times 10^5\,Pa = 750\,mmHg$

$1\,atm = 760\,mmHg = 1.013 \times 10^5\,Pa = 101.3\,kPa$

Worked example 14.1.2

CONVERTING PRESSURE UNITS

Mount Everest is the highest mountain on Earth.

a The atmospheric pressure at the top of Mount Everest is 0.337 bar. What is the pressure in kilopascals (kPa)?

b The atmospheric pressure at the top of Mount Everest is 253 mmHg. What is the pressure in atmospheres (atm)?

c The atmospheric pressure at the top of Mount Everest is 0.333 atm. What is the pressure in kilopascals (kPa)?

d The atmospheric pressure at the top of Mount Everest is 253 mmHg. What is the pressure in bars?

Thinking	Working
a To convert bars to kilopascals, use the conversion relationship: 1 bar = 100 kPa To change bar to kPa, multiply the value by 100.	$0.337\,bar = 0.337 \times 100$ $= 33.7\,kPa$

b To convert millimetres of mercury to atmospheres, use the relationship: 1 atm = 760 mmHg To change mmHg to atm, divide the value by 760.	$253\,\text{mmHg} = \frac{253}{760}$ $= 0.333$ atm
c To convert atmospheres to kilopascals, use the conversion relationship: 1 atm = 101.3 kPa To change atm to kPa, multiply the value by 101.3.	$0.333\,\text{atm} = 0.333 \times 101.3$ $= 33.7\,\text{kPa}$
d This can be done in two steps. First, convert millimetres of mercury to atmospheres. Use the conversion relationship: 760 mmHg = 1 atm To change mmHg to atm, divide the value by 760. Keep the answer in your calculator and proceed to the next step. Next, convert atmospheres to bar. Use the conversion relationship: 1 atm = 1.013 bar To change atm to bar, multiply the quotient from the previous step by 1.013.	$253\,\text{mmHg} = \frac{253}{760}$ $253\,\text{mmHg} = \frac{253}{760} \times 1.013$ $= 0.337$ bar

Worked example: Try yourself 14.1.2

CONVERTING PRESSURE UNITS

Cyclone Yasi was one of the biggest cyclones in Australian history.

a The atmospheric pressure in the eye of Cyclone Yasi was measured as 0.902 bar. What was the pressure in kilopascals (kPa)?

b What was the pressure in the eye of Cyclone Yasi in atmospheres (atm) if it is known to be 677 mmHg?

c If the atmospheric pressure in the eye of Cyclone Yasi was 0.891 atm, what was the pressure in kilopascals (kPa)?

d The atmospheric pressure in the eye of Cyclone Yasi was 677 mmHg. What was the pressure in bars?

CHEMISTRY IN ACTION

Decompression chambers

Scuba diving and water pressure

If you swim at the water's surface, your body experiences a pressure of about 1 atm, due to the surrounding air. Below the surface, your body experiences an additional pressure due to the water. This additional pressure amounts to about 1 atm for every 10 m of depth. Therefore, at 20 m the pressure on your body is about 3 atm.

As the pressure on your body increases, the volume of your body cavities such as your lungs and inner ears decrease. This squeezing effect makes diving well below the water's surface without scuba equipment very uncomfortable. Scuba equipment overcomes this problem by supplying air from tanks to the mouth at the same pressure as that produced by the underwater environment.

Scuba diving and gas solubility

As the pressure in a diver's lungs increases during a dive, more gas dissolves in the blood. Nitrogen (N_2) is one of these gases. When a diver ascends, the pressure drops, the nitrogen becomes less soluble in the blood and so comes out of solution. If a diver ascends too quickly, the rapid pressure drop causes the nitrogen to come out of the blood as tiny bubbles (Figure 14.1.14a). This is similar to the bubbles of carbon dioxide you observe when you open a bottle of soft drink.

These bubbles cause pain in joints and muscles. If they form in the spinal cord, brain or lungs, they can cause paralysis or death. Divers suffering from this effect (the bends) are put in a decompression chamber, like the one shown in Figure 14.1.14b. The chamber increases the pressure surrounding the diver's body, forcing any nitrogen bubbles to dissolve in the blood, and then slowly reduces the pressure back to 1 atm over an extended period of time.

Hyperbaric oxygen therapy (HBOT) is a medical treatment in which patients breathe pure oxygen while inside a decompression chamber at a pressure higher than 1 atm. Its medical uses include improved wound healing by reduction of swelling, infection control and the stimulation of new blood vessel growth. The hyperbaric unit at the Alfred Hospital in Melbourne is equipped with three pressurisable hyperbaric rooms used for treating the bends, ulcers, soft tissue infections, carbon monoxide poisoning and wounds that won't heal such as those resulting from diabetes or radiotherapy.

HBOT is also used by athletes for faster recovery from soft tissue injuries, to produce sharper performance and to maintain more intense training schedules. Tennis star Novak Djokovic uses a hyperbaric chamber to assist with recovery from long, exhausting matches. AFL clubs are interested in the use of hyperbaric oxygen therapy for treating soft tissue injuries. One of the earliest cases of hyperbaric oxygen treatment in football was for an ankle injury sustained by Carlton midfielder Fraser Brown in the 1995 preliminary final against North Melbourne. Every morning, in the week leading up to the Grand Final, he spent an hour in a chamber in the hyperbaric unit at the Alfred Hospital. Eventually selected to play in the Grand Final, Brown made it through the game with the help of strapping and pain-killing injections, helping Carlton win the premiership.

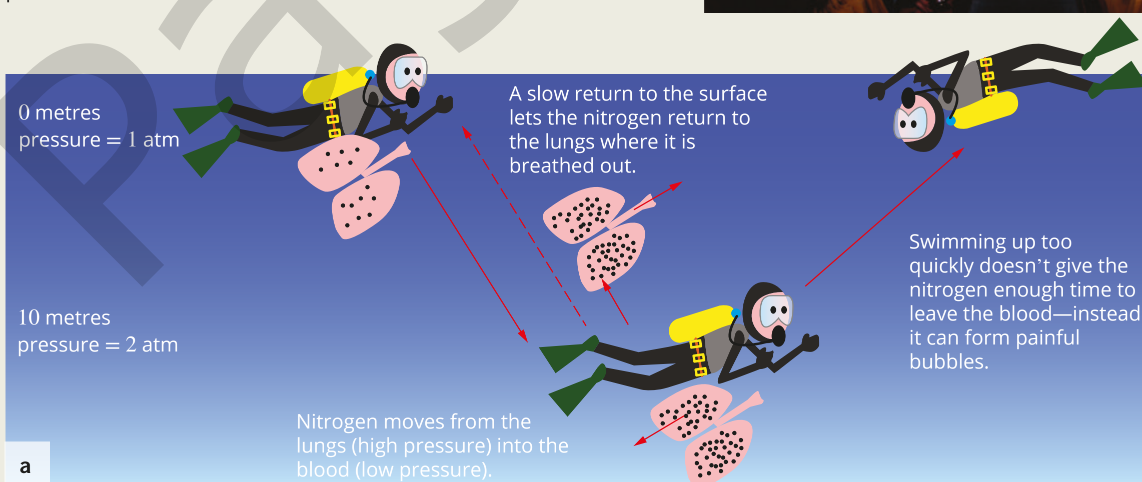


FIGURE 14.1.14 (a) Pressure differences during scuba diving can cause gases like nitrogen to become less soluble in the blood of scuba divers and lead to the bends. (b) A decompression chamber is used to treat divers with the bends.

EXTENSION

The relationship between volume, temperature and pressure

By performing experiments on gases, it is possible to establish some rules or laws that quantify the relationship between volume, pressure, temperature and the number of particles of gas. These relationships have become known as gas laws. The gas laws are beyond the scope of this course but are summarised below.

The relationship between gas volume and pressure

In 1662, Robert Boyle, an Irish chemist, showed by experiment that if the volume of a fixed amount of gas at constant temperature is halved, the pressure doubles. If the volume is tripled, the pressure drops to one third of its original value. Boyle concluded that, for a given amount of gas at constant temperature, the volume of the gas is inversely proportional to its pressure. The relationship between gas volume and pressure is known as Boyle's law.

Changing the volume of a fixed amount of gas at constant temperature causes a change in the pressure of the gas. The pressure of the gas in the syringe shown in Figure 14.1.15 increases as the plunger is pushed in and decreases as the plunger is pulled out.

This relationship is seen in the changing volume of a weather balloon as it rises to altitudes with much lower pressure than at ground level. A weather balloon filled with helium gas to a volume of 40 L at a pressure of 1 atm increases in volume to 200 L by the time it reaches an altitude with a pressure of 0.2 atm.

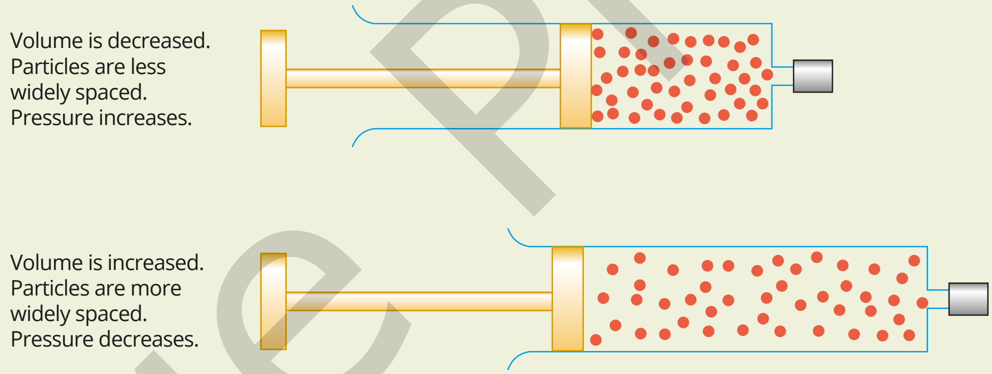


FIGURE 14.1.15 The pressure of the gas in the syringe is decreased when the plunger is pulled out. In this case, the syringe needle is capped, and no gas is allowed to enter or escape.

The relationship between gas volume and temperature

In 1787, Jacques Charles, experimentally determined that the volume of a fixed amount of gas is directly proportional to the absolute temperature, provided the pressure remains constant. The relationship between gas volume and temperature is known as Charles' law.

Table 14.1.3 shows the results of an experiment that investigates the relationship between temperature and the volume of a given amount of gas at constant pressure. The pressure on the plunger of the syringe is held constant. In this experiment, the gas in a syringe is heated slowly in an oven.

TABLE 14.1.3 Variation of volume with temperature

Temperature (°C)	20	40	60	80	100	120	140
Volume (mL)	60.0	64.1	68.2	72.3	76.4	80.5	84.6

EXTENSION *continued*

The graphs of these results, and the results of similar experiments with other gas samples, are linear, as shown in Figure 14.1.16. When the graph is extrapolated to a volume of 0 L, it crosses the temperature axis at −273.15°C. This has led scientists to develop a new temperature scale, known as the Kelvin scale or absolute temperature scale. On the Kelvin scale, each temperature increment is equal to one temperature increment on the Celsius scale, and 0°C is equal to 273.15 K.

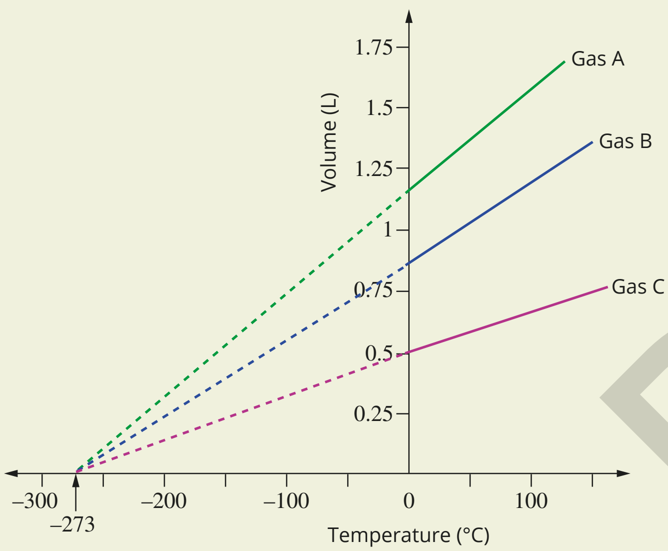


FIGURE 14.1.16 The variation of volume with temperature for fixed amount of three different gases at constant pressure

The relationship between temperature on the Celsius scale and temperature on the Kelvin scale is given by the equation:

$$T \text{ (in K)} = T \text{ (in °C)} + 273.15$$

The temperature 0 K (–273.15°C) is the lowest temperature theoretically possible. For this reason, 0 K is known as absolute zero. At this temperature, all molecules would have minimum kinetic energy and molecular motion will theoretically have stopped. The accepted convention for the value of absolute zero is –273.15°C.

When a weather balloon rises in the atmosphere, the pressure and temperature of the gas inside decrease. These changes have opposing effects on the volume of the balloon. The decrease in pressure causes the volume of the balloon to increase. To a lesser extent, the decrease in temperature causes the volume of the balloon to decrease. The effect is an overall increase in the volume of the balloon.

FIGURE 14.1.13 When a weather balloon rises in the atmosphere, its volume increases as a result of the decrease in atmospheric pressure.

14.1 Review

SUMMARY

- Gases have relatively low density, they are easily compressed and different gases mix rapidly together.
- Gas properties can be explained by the kinetic molecular theory.
- According to the kinetic molecular theory:
  - the volume of the particles in a gas is very small compared with the distance between the particles
  - the average kinetic energy of the particles in a gas is proportional to its temperature
  - gas particles are in rapid random motion, colliding with each other and the container wall
  - the forces between particles in a sample of gas are negligible.
- Gases that behave according to the kinetic molecular theory are called ideal gases.
- Diffusion is the process whereby gases in a mixture spread out to uniformly fill the total volume available.
- Pressure is defined as the force per unit area.
- The pressure exerted by a gas is caused by the collisions of gas particles against the wall of the container.
- To convert between different volume units, use these relationships:
  $1\,\text{L} = 1000\,\text{mL} = 1000\,\text{cm}^3$
  $1\,\text{m}^3 = 1 \times 10^6\,\text{cm}^3 = 1 \times 10^6\,\text{mL} = 1000\,\text{L}$
- To convert between different pressure units, use these relationships:
  $1\,\text{bar} = 100\,\text{kPa} = 1.00 \times 10^5\,\text{Pa} = 750\,\text{mmHg}$
  $1\,\text{atm} = 760\,\text{mmHg} = 1.013 \times 10^5\,\text{Pa} = 101.3\,\text{kPa} = 1.013\,\text{bar}$

KEY QUESTIONS

1 Use the kinetic molecular theory to explain the following observed properties of gases.
  a Gases occupy all the available space in a container.
  b Gases can be easily compressed compared with their corresponding liquid forms.
  c A given volume of a gaseous substance weighs less than the same volume of the substance in the liquid state.
  d Gases readily mix together.
  e The total pressure of a mixture of gases is equal to the sum of the pressures exerted by each of the gases in the mixture.

2 Use the ideas of the kinetic molecular theory of gases to explain the following observations.
  a Tyre manufacturers recommend a maximum pressure for tyres.
  b The pressure in a car's tyres will increase if a long distance is travelled on a hot day.
  c You can smell dinner cooking as you enter your house.
  d A balloon will burst if you blow it up too much.

3 Four gases and their molar masses are listed below. If the gases are at the same temperature, sort them to show their rate of diffusion from fastest to slowest.
  - argon ($39.95\,\text{g mol}^{-1}$)
  - helium ($4.003\,\text{g mol}^{-1}$)
  - krypton ($83.80\,\text{g mol}^{-1}$)
  - neon ($20.18\,\text{g mol}^{-1}$)

4 In the kinetic molecular theory, pressure is described as the force per unit area of surface. Explain what happens to the pressure in each of the following situations.
  a The temperature of a filled aerosol can is increased.
  b A gas in a syringe is compressed.

5 Convert each of the following pressures to the units specified.
  a 140 kPa to Pa
  b 92 000 Pa to kPa
  c 4.24 atm to mmHg and Pa
  d 120 kPa to mmHg, atm and bar

6 Convert the following volumes to the unit specified.
  a 2 L to mL
  b 4.5 L to m^3
  c 2250 mL to L
  d 120 mL to L

14.2 Molar volume of a gas

In 1811, the Italian scientist Amedeo Avogadro proposed that, at the same temperature and pressure, equal volumes of all gases contain equal numbers of particles. This became known as **Avogadro's law**. The law can be summarised by the relationship:

$$V \propto n$$

The volume, V, occupied by a gas depends directly on the amount of gas, n, in moles, at a constant pressure and temperature.

This relationship is shown in Figure 14.2.1. Both syringes show a gas at a constant temperature and pressure. The volume doubles with twice the number of molecules of gas in the syringe.

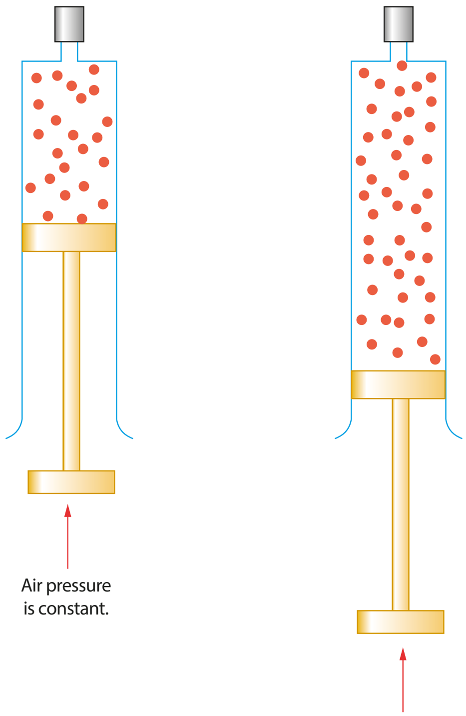


FIGURE 14.2.1 When the amount of gas in the syringe is doubled, the volume doubles, provided the pressure on the plunger and the temperature of the gas remain constant.

> At the same temperature and pressure, equal volumes of all gases contain equal numbers of particles. The law can be summarised by the relationship:
> $$V \propto n$$

CHEMFILE

Amedeo Avogadro

Avogadro's full name was Lorenzo Romano Amedeo Carlo Avogadro and his title was Count of Quaregna and Cerrato. Born in Italy in 1776, he worked for three years as a lawyer before leaving the legal profession to study physics and mathematics. This path eventually led him to formulate his hypothesis, which is now known as Avogadro's law: under identical conditions of temperature and pressure, equal volumes of gases contain an equal number of particles. It was not until five years after his death that the Italian chemist Stanislao Cannizzaro presented Avogadro's work and he finally received credit for it.

FIGURE 14.2.2 Amedeo Avogadro (1776–1856)

If the amount of gas is fixed at 1 mole, the volume the gas occupies will depend almost entirely on its temperature and pressure. This volume is defined as the **molar volume**, V_m, of a gas. Molar volume is the amount of space, or volume, occupied by 1 mole of any gas.

The volume of 1 mol of gas, V_m, is equal to its total volume, V, divided by the number of moles, n, of gas present. This can be represented by the relationship:

$$V_m = \frac{V}{n} \quad \text{(at a given temperature and pressure)}$$

Or, by rearranging this expression:

$$n = \frac{V}{V_m} \quad \text{(at a given temperature and pressure)}$$

The molar volume of a gas varies with temperature and pressure. However, molar volume does not vary with the identity of the gas. For example, one mole of neon gas will occupy the same volume as one mole of nitrogen and one mole of ozone gas at the same temperature and pressure (see Figure 14.2.3).

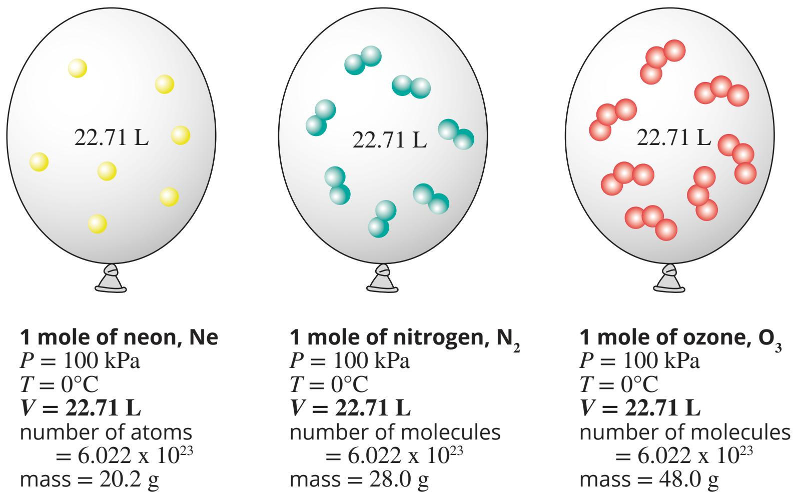


FIGURE 14.2.3 1 mole of any gas occupies a volume of 22.71 L at STP.

Standard conditions

Standard temperature and pressure (STP) refers to a temperature of 0°C (273.15 K) and a pressure of 100 kPa. Table 14.2.1 shows the molar volume of an ideal gas and of helium at STP.

TABLE 14.2.1 Molar volume at STP

Gas	Formula	Molar volume at STP (L mol^{-1})
Ideal gas	–	22.71
Helium	He	22.69

An ideal gas is a theoretical gas composed of particles that do not interact except during elastic collisions. At STP, most gases behave very like an ideal gas and, therefore, have a molar volume very like that of an ideal gas.

It is usual to assume that the molar volume of a gas is 22.71 L mol^{-1} at STP. From this value the amount, in moles, of a gas given its volume at STP can be calculated.

From the table above it can be seen that the volumes of real gases vary slightly at STP. The stronger the intermolecular forces between the gas molecules, the greater their volume varies from that of an ideal gas at STP.

The observation that gases only approximate ideal behaviour suggests that they do not behave exactly according to the kinetic molecular theory. Generally, a gas behaves more like an ideal gas at higher temperature and lower pressure. At STP, most gases behave like an ideal gas.

Although the molar volume of a gas at STP is commonly used, it is important to remember that one mole of all gases at the same temperature and pressure will occupy a fixed volume, regardless of the particular temperature and pressure. This is Avogadro's law.

V_m = 22.71 L mol^{-1} at STP (0°C or 273.15 K and 100 kPa)

Worked example 14.2.1

CALCULATING THE VOLUME OF A GAS FROM ITS AMOUNT (IN MOL)

Calculate the volume, in L, occupied by 0.24 mol of nitrogen gas at STP. Assume that nitrogen behaves like an ideal gas.	
Thinking	**Working**
Rearrange $n = \frac{V}{V_m}$ to make volume the subject.	$n = \frac{V}{V_m}$ $V = n \times V_m$
Substitute in the known values where $V_m = 22.71\ \text{L mol}^{-1}$ (at STP) and solve.	$V_{(STP)} = n \times V_m$ $= 0.24 \times 22.71$ $= 5.450\ \text{L}$
Consider the units and significant figures. The answer should be given to the smallest number of significant figures in the measurement.	$V_{(STP)} = 5.5\ \text{L}$

Worked example: Try yourself 14.2.1

CALCULATING THE VOLUME OF A GAS FROM ITS AMOUNT (IN MOL)

Calculate the volume, in L, occupied by 3.50 mol of oxygen gas at STP. Assume that oxygen behaves like an ideal gas.

Worked example 14.2.2

CALCULATING THE AMOUNT (IN MOL) OF A GAS FROM ITS VOLUME

Calculate the amount (in mol) of nitrogen gas in a volume of 6.1 L at STP. Assume that nitrogen behaves like an ideal gas.	
Thinking	**Working**
Use $n = \frac{V}{V_m}$ where $V_m = 22.71\ \text{L mol}^{-1}$ (at STP)	$n = \frac{V}{V_m}$ $= \frac{6.1}{22.71}$ $= 0.2686\ \text{mol}$
Consider the units and significant figures. The answer should be given to the smallest number of significant figures in the measurement.	$n = 0.27\ \text{mol}$

Worked example: Try yourself 14.2.2

CALCULATING THE AMOUNT (IN MOL) OF A GAS FROM ITS VOLUME

Calculate the amount (in mol) of oxygen gas in a volume of 3.5 L measured at STP. Assume that oxygen behaves like an ideal gas.

14.2 Review

SUMMARY

- Standard temperature and pressure (STP) refers to a temperature of 0°C (273 K) and a pressure of 100 kPa.
- The molar volume, V_m, of a gas is the volume occupied by 1 mol of gas at a given temperature and pressure:

$$n = \frac{V}{V_m}$$

- The value of V_m is 22.71 L mol^{-1} at STP.
- Avogadro's law states that equal volumes of gas at the same temperature and pressure contain the same number of particles.

KEY QUESTIONS

1 Calculate the volume of the following gases at STP.
   a 1.4 mol of chlorine (Cl_2)
   b 1.0×10^{-3} mol of hydrogen (H_2)
   c 1.4 g of nitrogen (N_2)

2 Calculate the number of moles of the following gas samples. All volumes are measured at STP.
   a 2.80 L of neon (Ne)
   b 50.0 L of oxygen (O_2)
   c 140 mL of carbon dioxide (CO_2)

3 Which one of the following is the approximate volume of 1.4 mol of chlorine gas, Cl_2, at STP?
   A 63.59 L
   B 16.22 L
   C 0.0616 L
   D 32 L

14.3 Calculations involving reactions with gases

In this section you will learn how you can combine the mole concept with your understanding of molar volume to calculate the mass or volume of a gas in a chemical reaction. This is useful in many situations, including determining the volume of carbon dioxide gas produced by fuels or the mass of water vapour produced in a combustion reaction.

Calculations that involve the use of the mole concept, combined with an understanding of chemical equations, are called stoichiometric calculations which were introduced in Chapter 11. Stoichiometry is the study of ratios of moles of substances. Stoichiometric calculations are based on the law of conservation of mass.

Another way of expressing this is that, in a chemical reaction, atoms are neither created nor destroyed. Consequently, given the amount of one substance involved in a chemical reaction and a balanced equation for the reaction, you can calculate the amounts of all other substances involved.

> In a chemical reaction, the total mass of all products is equal to the total mass of all reactants.

EQUATIONS AND REACTING AMOUNTS

Consider the equation for the reaction that occurs when methane (CH_4) burns in oxygen:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

The coefficients used to balance the equations show the ratios between the reactants and products involved in the reaction. The equation indicates that 1 mole of $CH_4(g)$ reacts with 2 moles of $O_2(g)$ to form 1 mole of $CO_2(g)$ and 2 moles of $H_2O(g)$. In more general terms, the amount of oxygen used will always be double the amount of methane used, double the amount of carbon dioxide produced and the same as the amount of water vapour produced.

$$\frac{n(O_2)}{n(CH_4)} = \frac{2}{1}, \ \frac{n(CO_2)}{n(O_2)} = \frac{1}{2} \text{ and } \frac{n(H_2O)}{n(O_2)} = \frac{2}{2} = 1$$

In general, for stoichiometric calculations you will be told, or you will be able to work out, the number of moles of one chemical in the reaction (the 'known'). You will then need to calculate the number of moles of another chemical (the 'unknown').

You can write the relationship between the known and unknown chemicals using ratios:

$$\frac{n(\text{unknown chemical})}{n(\text{known chemical})} = \frac{\text{coefficient of unknown chemical}}{\text{coefficient of known chemical}}$$

Worked example 14.3.1

USING MOLE RATIOS

How many moles of carbon dioxide are generated when 5.5 moles of propane (C_3H_8) are burned completely in oxygen?	
Thinking	**Working**
Write a balanced equation for the reaction.	$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$
Note the number of moles of the known substance.	$n(C_3H_8) = 5.5\,mol$
Write a mole ratio for: $\frac{\text{coefficient of unknown}}{\text{coefficient of known}}$	$\frac{n(CO_2)}{n(C_3H_8)} = \frac{3}{1}$
Calculate the number of moles of the unknown substance using: n(unknown) = mole ratio × n(known)	$n(CO_2) = \frac{3}{1} \times 5.5$ $= 16.5\,mol$

When carrying out any stoichiometric calculations, you must always clearly state the mole ratio for the reaction you are working with.

Worked example: Try yourself 14.3.1

USING MOLE RATIOS

How many moles of carbon dioxide are generated when 2.55×10^3 moles of butane (C_4H_{10}) are burned completely in oxygen?

MASS–MASS STOICHIOMETRY

Calculations can require you to start and finish with masses rather than moles, as this is how quantities of chemicals are often measured.

Stoichiometric calculations generally follow the same pattern. The number of moles of a 'known' substance is calculated from data that is given to you, the mole ratios in the equation are used to find the number of moles of the 'unknown' substance, and the desired quantity of the unknown substance is then calculated.

Figure 14.3.1 shows the equations used to determine the volume, moles or mass of reactants or products in a chemical reaction involving gases. To determine these quantities for a given reaction, one must start with a balanced chemical equation.

- To calculate the number of moles from the mass of a substance:

  $$\text{moles}\,(n) = \frac{\text{mass in g}\,(m)}{\text{molar mass}\,(M)}$$

- To calculate the mass from the number of moles, rearrange this relationship as:

  mass in g (m) = number of moles (n) × molar mass (M)

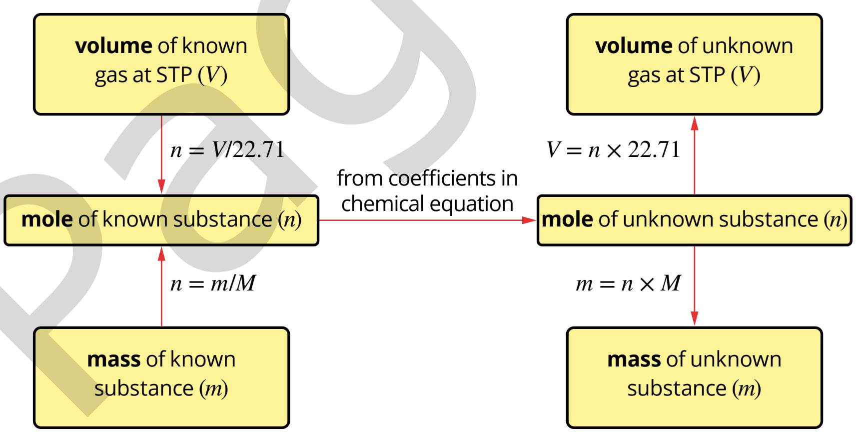


FIGURE 14.3.1 Most gas stoichiometric calculations follow the steps shown in the flow chart. Calculating the number of moles and using a mole ratio from a balanced chemical equation are always central to any stoichiometric calculation.

When carrying out stoichiometric calculations, always keep the answer in your calculator and proceed to the next step. Only round to the correct number of significant figures when finished.

MASS–VOLUME STOICHIOMETRY

Some stoichiometric calculations require you to determine the volume of a gas that reacts with, or is produced from, a given mass of a substance. For these calculations, you need to determine the number of moles of the substance from its mass, and use the mole ratio from the balanced chemical equation.

The volume of the gas, in litres, is determined from its amount, in moles, at standard temperature of 0°C and 100 kPa (STP). The molar volume equation is used:

$$n = \frac{V}{V_m}$$

This formula can be rearranged to make volume the subject:

$$V = n \times V_m$$

Remember that, at STP, the accepted volume of one mole of any gas is 22.71 L

These calculations are based on the mole ratios of reactants and products. For example, the reaction that occurs when calcium carbonate is heated can be represented by the equation:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

Recall that the lack of a coefficient before a reactant or product in the equation indicates one mole. This balanced equation tells us that in any given reaction the amount, in mol, of calcium oxide produced is equal to the amount, in mol, of carbon dioxide produced. This is also equal to the amount, in mol, of calcium carbonate that decomposes.

Worked example 14.3.2

MASS–VOLUME STOICHIOMETRIC CALCULATIONS

A sample of calcium carbonate ($CaCO_3$) of mass 1.001 g, is heated until it has decomposed completely to form calcium oxide (CaO) and carbon dioxide (CO_2). Calculate the volume of carbon dioxide produced, measured at STP.

Thinking	Working
Write a balanced equation for the reaction.	$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
Identify the known quantity and the unknown.	Known: mass of $CaCO_3$ = 1.001 g Unknown: What is the volume of CO_2 at STP?
Calculate the number of moles, *n*, of the known substance using: $n = \frac{m}{M}$	$n(CaCO_3) = \frac{1.001}{100.09}$ $= 0.01000$ mol
Find the mole ratio: $\frac{\text{coefficient of unknown}}{\text{coefficient of known}}$	$\frac{n(CO_2)}{n(CaCO_3)} = \frac{1}{1}$
Calculate the number of moles of the unknown substance using: $n(\text{unknown}) = \text{mole ratio} \times n(\text{known})$	$n(CO_2) = n(CaCO_3)$ $= 0.01000$ mol
Calculate the volume of the unknown substance using: $V = n \times 22.71$	$V(CO_2) = n(CO_2) \times V_m$ $= 0.01000 \text{ mol} \times 22.71$ L $= 0.2271$ L $= 227$ mL

Worked example: Try yourself 14.3.2

MASS–VOLUME STOICHIOMETRIC CALCULATIONS

A sample of calcium carbonate ($CaCO_3$) is heated until it has decomposed completely to form calcium oxide (CaO) and carbon dioxide (CO_2). A mass of 3.00 g of calcium oxide is produced. Calculate the volume of carbon dioxide produced, measured at STP. Give your answer to 3 significant figures.

GAS VOLUME–VOLUME CALCULATIONS

For chemical reactions where both the reactants and products are in the gaseous state, it is often convenient to measure volumes, rather than masses.

For example, the reaction between hydrogen and chlorine can be represented by the equation:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

This equation tells us that when equal numbers of moles of hydrogen gas and chlorine gas react, twice as many moles of hydrogen chloride gas are produced.

Avogadro's law tells us that equal amounts, in moles, of all gases occupy equal volumes measured at the same temperature and pressure.

Therefore, the mole ratios in the balanced equation become volume ratios at the same temperature and pressure. In the above reaction, this means that when equal volumes of hydrogen gas and chlorine gas react they will produce twice the volume of hydrogen chloride.

Worked example 14.3.3

GAS VOLUME–VOLUME CALCULATIONS

Methane gas (CH_4) is burned in a gas stove according to the following equation:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H2O(g)$$

If 50 mL of methane is burned, calculate the volume of O_2 gas required for complete combustion of the methane under constant temperature and pressure conditions.

Thinking	Working
Use the balanced equation to find the mole ratio of the two gases involved.	1 mol of CH_4 gas reacts with 2 mol of O_2 gas.
The temperature and pressure are constant, so volume ratios are the same as mole ratios.	1 volume of CH_4 reacts with 2 volumes of O_2 gas, so 50 mL of CH_4 reacts with 100 mL of O_2.

Worked example: Try yourself 14.3.3

GAS VOLUME–VOLUME CALCULATIONS

Methane gas (CH_4) is burned in a gas stove according to the following equation:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

If 50 mL of methane is burned in air, calculate the volume of CO_2 gas produced under constant temperature and pressure conditions.

EXTENSION

Calculations involving excess reactants

Stoichiometry calculations become more complex if the reactants are not present in their stoichiometric ratio. In these cases, you must determine which reactant is completely consumed in the reaction, the limiting reactant, and which one is present in excess. The amount of limiting reactant determines how much product is formed.

Worked example 14.3.4 introduces a strategy that can be used to determine the limiting reactant in a reaction.

Worked example 14.3.4

EXCESS REACTANT CALCULATIONS

A gaseous mixture of 25.0 g of hydrogen gas and 100.0 g of oxygen gas are mixed and ignited. The water produced is collected and weighed. What is the expected mass of water produced? The equation for the reaction is:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$$

a Which reactant is the limiting reactant?

b What is the mass of water vapour formed?

Thinking	Working
a Calculate the number of moles of each reactant using $n = \frac{m}{M}$ or $n = \frac{V}{V_m}$ as appropriate.	$n(H_2) = \frac{m}{M} = \frac{25.0}{2.0} = 12.5\text{ mol}$ $n(O_2) = \frac{m}{M} = \frac{100.0}{32.0} = 3.13\text{ mol}$
Use the coefficients of the equation to find the limiting reactant.	The equation shows 2 mol of H_2 reacts with 1 mol of O_2. So to react all of the O_2 will require $2 \times n(O_2)$ of $H_2 = 2 \times 3.13 = 6.26\text{ mol}$ As there is 12.5 mol of H_2, the H_2 is in excess. The O_2 is the limiting reactant (it will be completely consumed).
b Find the mole ratio using: $\frac{\text{coefficient of unknown}}{\text{coefficient of known}}$ The limiting reactant is the known substance.	$\frac{n(H_2O)}{n(O_2)} = \frac{2}{1}$
Calculate the number of moles of the unknown substance using: $n(\text{unknown}) = \text{mole ratio} \times n(\text{known})$	$n(H_2O) = 2 \times 3.13$ $= 6.26\text{ mol}$
Calculate the required quantity of the unknown using $m = n \times M$ or $V = n$ as appropriate.	$m(H_2O) = 6.26 \times 18.016$ $= 113\text{ g}$

Worked example: Try yourself 14.3.4

EXCESS REACTANT CALCULATIONS

Calculate the volume of carbon dioxide, in L, produced when 65.0 g of butane is burned completely in 200 L of oxygen. The gas volume is measured at STP. The equation for the reaction is:

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$$

a Which reactant is the limiting reactant?

b What is the volume of carbon dioxide formed?

In these types of problems, remember to always use the number of moles of the limiting reactant to determine the amount of product that will be formed.

14.3 Review

SUMMARY

- A balanced equation shows the ratio of the amount, in moles, of reactants and products in the reaction.
- Stoichiometric calculations follow the general steps:
  1 Calculate the amount, in moles, of a known substance from the data given.
  Use $n = \frac{m}{M}$, or $n = \frac{V}{V_m}$
  2 Use the mole ratio from a balanced chemical equation to determine the amount, in moles, of the unknown substance.

$$\frac{n(\text{unknown chemical})}{n(\text{known chemical})} = \frac{\text{coefficient of unknown chemical}}{\text{coefficient of known chemical}}$$

  3 Find the desired quantity of the unknown substance from its amount, in moles, using $m = n \times M$ or $V = n \times 22.71$
- Stoichiometric calculations can be used to calculate the mass of substances involved in chemical reactions.
- The mole ratio in a balanced equation is also a volume ratio if all reactants and products are in the gaseous state and the temperature and pressure are kept constant.
- The volume of gases (at standard temperature and pressure) involved in a chemical reaction can also be calculated using the mole concept.

KEY QUESTIONS

1 Methanol burns in air according to the equation:

$$2CH_3OH(g) + 3O_2(g) \rightarrow 2CO_2(g) + 4H_2O(g)$$

Complete the following mathematical relationships:

a $\frac{n(CH_3OH)}{n(O_2)} =$

b $\frac{n(O_2)}{n(H_2O)} =$

c $\frac{n(CH_3OH)}{n(CO_2)} =$

2 Create a flow chart for completing mass–mass stoichiometric calculations by placing the steps in the correct order:
- Identify the known and unknown substances in the question.
- Use mole ratios from the equation to calculate the amount of the unknown substance.
- Calculate the mass of the unknown substance using $m = n \times M$.
- Write a balanced equation for the reaction.
- Calculate the amount, in mol, of the known substance using $n = \frac{m}{M}$.

3 Ammonia (NH_3) is produced by the reaction between hydrogen gas (H_2) and nitrogen gas (N_2) according to the reaction:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

Assuming complete reaction, what mass of ammonia would be produced when 12.0 L of hydrogen gas reacts with excess nitrogen gas at STP?

4 Glucose ($C_6H_{12}O_6$) ferments in an aqueous solution according to the equation:

$$C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

Express your answers to the following questions to 3 significant figures.

a If 3.604 g of glucose ferments, what is the amount in moles?

b Determine the number of moles of CO_2 that are produced when this number of moles of glucose ferments.

c Determine the volume of carbon dioxide (CO_2) measured at STP, that is produced.

5 Calculate the volume of oxygen gas (O_2) measured at STP, that is prepared when 7.35 g of potassium chlorate is decomposed by strong heat according to the equation below:

$$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$$

Express your answer to 3 significant figures.

6 Methane will burn in excess oxygen according to the equation:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

This reaction produces 5.00 L of carbon dioxide (CO_2) at STP. Calculate the mass of water vapour produced. Express your answer to 3 significant figures.

Chapter review

14

KEY TERMS

Avogadro's law
bar
diffusion
elastic collision
ideal gas
kinetic energy
kinetic energy distribution diagram
kinetic molecular theory
Maxwell–Boltzmann distribution graph
molar volume
partial pressure
pascal
pressure
standard temperature and pressure (STP)
volume

Introducing gases

1 Which one of the following volumes is equal to 4.5 L?
A 4.5×10^2 mL
B 4.5×10^3 mL
C 4.5×10^{-3} mL
D 4.5×10 mL

2 Which one of the following best describes the effect of an increase in temperature on gas particles?
A Both the average kinetic energy and the average speed of the particles increase.
B Both the average kinetic energy and the average speed of the particles decrease.
C The average kinetic energy of the particles increases and the average speed of the particles decreases.
D The average kinetic energy of the particles decreases and the average speed of the particles increases.

3 Which one of the following would you expect to happen when a container of a smelly gas is opened in the corner of a small, closed room?
A The smell of the gas will fill the room in a relatively short time.
B The smell of the gas will be confined to the corner of the room were the container is found.
C The smell of the gas will travel across the room and the corner where the container was opened no longer smells.
D The smell of the gas will fill half of the room and will travel no further.

4 Select the correct answers from the pairs of words to complete the paragraph about gases.
The volume occupied by the atoms or molecules in a gas is much *smaller/larger* than the total volume occupied by the gas. The particles move in rapid, *straight-line/curved* paths and collide with each other and with the walls of the container. The forces between particles are extremely *weak/strong*. The collisions between particles are *elastic/rigid*. The average kinetic energy of the particles is *directly/inversely* proportional to the temperature of the gas, in units of *K/°C*.

5 Use the kinetic theory of gases to explain why:
a the pressure of a gas increases if its volume is reduced at constant temperature
b the pressure of a gas decreases if its temperature is lowered at a constant volume
c in a mixture of gases, the total pressure is the sum of the partial pressure of each gas
d the pressure of a gas, held at constant volume and temperature, will increase if more gas is added to the container.

6 **a** If a container of gas is opened and some of the gas escapes, what happens to the pressure of the remaining gas in the container?
b Use the kinetic molecular theory to explain what happens to the gas pressure in part **a**.

7 Which one of the following explains why an inflated balloon might pop when its temperature is increased?
A The mass of gas inside the balloon increases.
B The pressure of gas inside the balloon increases.
C The number of molecules inside the balloon increases.
D The pressure of gas outside the balloon decreases.

Molar volume of gases

8 Determine the amount (in mol) of oxygen gas in a volume of 500 mL measured at STP. Assume the gas behaves like an ideal gas.

9 Determine the volume occupied by 0.1100 g of carbon dioxide gas (CO_2) at STP. Assume that carbon dioxide behaves like an ideal gas. Give your answer to 3 significant figures.

Calculations involving reactions with gases

10 Which one of the following is the mass of 50 L of oxygen gas (O_2) measured at STP?
A 7.05 g
B 2.20 g
C 70.5 g
D 32.0 g

11 Use the molar volume of a gas at STP to find the:
  a volume occupied by 8.0 g of oxygen
  b mass of nitrogen dioxide present in 10 L.

12 Carbon dioxide gas is a product of the complete combustion of fuels.
  a Calculate the mass of 1.00 mol of carbon dioxide.
  b What is the volume occupied by 1.00 mol of carbon dioxide at STP?
  c Given that density (d) is defined as $\frac{\text{mass}}{\text{volume}}$, calculate the density of carbon dioxide at STP in $g\,L^{-1}$.

13 A room has a volume of 220 m^3.
  a Calculate the amount, in moles, of air particles in the room at STP.
  b Assume that 20% of the molecules in the air are oxygen molecules and the remaining molecules are nitrogen. Calculate the mass of nitrogen in the room.

14 The ethanol produced by the fermentation of glucose is used as a biochemical fuel or biofuel. Fermentation of glucose produces ethanol and carbon dioxide according to the following equation:

$$C_6H_{12}O_6(aq) \xrightarrow{\text{yeast}} 2C_2H_5OH(aq) + 2CO_2(g)$$

Calculate the mass of carbon dioxide produced by the fermentation of 80.0 g of glucose.

15 Large quantities of coal are burned in Australia to generate electricity, in the process generating significant amounts of the greenhouse gas carbon dioxide. The equation for this combustion reaction is:

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

Determine the mass of carbon dioxide produced by the combustion of 1.0 tonne (10^6 g) of coal, assuming that the coal is pure carbon.

16 Propane (C_3H_8) burns in oxygen according to the equation:

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

Calculate the volume of:
  i oxygen at STP used
  ii carbon dioxide at STP produced

when the following masses of propane react completely with excess oxygen.
  a 22 g
  b 5.0 g
  c 0.145 g

Connecting the main ideas

17 Which of the following statements is not correct?
  A At the same temperature, a sample of CO_2 would exert a greater pressure than the same amount of helium molecules.
  B The measured air pressure is the sum of all the individual gas pressures in the sample of air.
  C The pressure of a fixed amount of gas is independent of the identity of the gas.
  D As the temperature of a sample of gas increases in a fixed volume container, the pressure of the sample will increase.

18 Suppose that 350 kg of iron(II) oxide (Fe_2O_3) reacts with excess carbon monoxide (CO) according to the equation.

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$$

  a Determine the number of moles of Fe_2O_3.
  b Determine the mole ratio of CO_2 to Fe_2O_3 and use this to calculate the number of moles of carbon dioxide produced.
  c Determine the volume of carbon dioxide (CO_2) measured at STP.

19 Consider two containers of equal size. One contains oxygen and the other carbon dioxide. Both containers are at 23°C and at a pressure of 1.0 atm. Answer each of the following questions about the two gases and give a reason for your answers.
  a Which of the two samples of gas contains more molecules?
  b Which of the two samples of gas contains the greater number of atoms?
  c Which of the two gases has the greater density?

20 Nitric acid (HNO_3) is decomposed by light according to the equation:

$$4HNO_3(aq) \rightarrow 4NO_2(g) + 2H_2O(l) + O_2(g)$$

  a From the equation, determine the molar ratio of NO_2 to HNO_3. If 0.500 mol of nitric acid is decomposed, how many moles of NO_2 gas are produced?
  b From the equation, determine the mole ratio of O_2 to HNO_3. If 0.500 mol of nitric acid is decomposed, how many moles of O_2 gas are produced?
  c Determine the total volume of gases produced at STP if 0.500 mol of NO_2 and 0.125 mol of O_2 are produced in the reaction.

21 Copper is extracted from chalcopyrite ($CuFeS_2$) according to the equation below.

$$2CuFeS_2(s) + 5O_2(g) + 2SiO_2(s) \rightarrow 2Cu(l) + 4SO_2(g) + 2FeSiO_3(l)$$

a If 350 L of oxygen gas measured at STP reacts, how many moles is this?

b Determine the number of moles of chalcopyrite, that would react with 350 L of O_2, measured at STP.

c Calculate the mass, in kilograms, of chalcopyrite that would react.

23 Hot-air balloons rise because hot air is less dense than cooler air.

a Explain why the density of a gas inside a hot-air balloon decreases as the temperature rises.

b A Chemistry student was explaining to a friend that as air is heated, the added heat causes the kinetic energy of all the air molecules to increase.
Comment critically on this statement, identifying any inconsistencies and making appropriate corrections.

c Use the kinetic molecular theory of gases to explain why the pressure of a gas decreases if its temperature is lowered at a constant volume.